Homework 7

Due Wednesday, 4 November 2026

1. An experimenter has prepared a drug dosage level that she claims will induce sleep for 80% of people suffering from insomnia. After examining the dosage, another researcher feels that her claims regarding the effectiveness of the dosage are inflated. In an attempt to disprove her claim, they administer her prescribed dosage to 20 insomniacs, and observe Y, the number for which the drug dose induces sleep. They wish to test the hypothesis $\mathrm{H}_0 : p = 0.8$ versus the alternative $\mathrm{H}_a : p < 0.8$. Assume that the rejection region $\{y \leq 12\}$ is used.

   a. In the context of the scenario, what is a Type I error?
   b. Find the significance level, α, of the test.
   c. In the context of the scenario, what is a Type II error?
   d. Find the probability of a Type II error when $p = 0.6$.
   e. Find the probability of a Type II error when $p = 0.4$.
   f. Find the power when $p = 0.4$.
   g. Find the power when $p = 0.5$.
   h. Find the power when $p = 0.6$.
   i. Find the power when $p = 0.7$.
   j. Sketch a graph of the power function.
   k. What is the power when $p = 0.8$? Why is this the same as your answer in (b)?
   l. What would happen to the power at $p = 0.8$ if we change the rejection region to be $\{y \leq 13\}$? What would happen to the power at $p = 0.8$ if we change the rejection region to be $\{y \leq 11\}$? Briefly explain why each of these changes occur.

2. A sample of size 1 is taken from an exponential(β) distribution. To test $\mathrm{H}_0 : \beta = 1$ against $\mathrm{H}_a : \beta > 1$, a proposed test is

$$\phi(X) = \begin{cases} 1 & X > 2 \\ 0 & X \leq 2 \end{cases}$$

   a. Find the significance level of the test.
   b. Find the power function of the test.

3. Suppose Y is a random sample of size 1 from a population with density function

$$f(y|\theta) = \begin{cases} \theta y^{\theta-1} & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

   a. Sketch the power function of the test with rejection region: $Y > 0.5$.
   b. The hypotheses are $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$. What is the value of θ_0 if the significance level of the test specified in (a) is 0.10?
   c. Suppose a random sample of $n = 7$ observations is taken from this population to test the hypotheses $\mathrm{H}_0 : \theta = 2$ versus $\mathrm{H}_a : \theta > 2$. As a decision rule, the experimenter plans to record X, the number of ys that exceed 0.9, and reject if $X \geq 4$. That is, the experimenter plans to use the test:

$$\phi(X) = \begin{cases} 1 & X \geq 4 \\ 0 & X < 4 \end{cases} .$$

   What is the probability of making a Type I error?

4. Suppose that X is one observation from the distribution with pdf $f(x|\lambda) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$. Consider testing

$$\mathrm{H}_0 : \lambda = 1 \qquad \text{versus} \qquad \mathrm{H}_1 : \lambda \neq 1$$

Let ϕ be the critical function

$$\phi(X) = \begin{cases} 1 & x < \frac{1}{2} \text{ or } x > 2 \\ 0 & \text{else} \end{cases}$$

a. Find the power function, $power(\lambda)$, of the test corresponding to ϕ.
b. What is the significance level of the test determined by ϕ?

5. A statistics instructor wants to figure out how long it will take her students, on average, to complete an exam. She believes that completion time should follow an exponential distribution with mean β. She collects data from 10 randomly selected students, and the class meets for 50 minutes.

a. What are the appropriate hypotheses in this case? Explain your reasoning.
b. The instructor opts to use a significance level of 0.10. In the context of the scenario, what does this mean?
c. The test the instructor uses has the form

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & \sum_i X_i \geq \frac{\lambda_0}{2} \chi^2_{20,0.10} \\ 0 & \text{elsewhere} \end{cases}$$

The completion times from the 10 randomly selected students are:

$$19, 29, 30, 35, 42, 44, 49, 58, 61, 186$$

What should the instructor conclude?
d. Another instructor thinks that a significance level of 0.01 would be a better choice. Which significance value would you recommend to the instructor? Explain your reasoning.
e. If you were writing an exam, would a similar analysis and its results be useful to you? Why or why not?